EEL 4657 - Linear Control Systems Final Project Report

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Course Project Sections: Modified Example 9.4.1, Example 9.6.1, and Skill Assessment 9.4.1

Software Used: MATLAB and Simulink

Report Summary. This report combines the hand-design discussion, controller implementation, MATLAB verification, and error analysis for the assigned Linear Control Systems final project. The main work includes a modified PID design based on Example 9.4.1, practical controller implementation using the method of Example 9.6.1, and a lag-lead compensator design for Skill Assessment 9.4.1.

1. Introduction

This project studies how classical compensator design changes the behavior of feedback control systems. The work was completed using the root-locus method first and then verified in MATLAB and Simulink. Three tasks were included in the assignment. The first task was to redesign Example 9.4.1 using the specifications assigned to my row. The second task was to implement the resulting PID controller using the circuit method from Example 9.6.1. The third task was to solve Skill Assessment 9.4.1, which required analysis of an uncompensated system and the design of a lag-lead compensator.

For my assigned row, the modified design of Example 9.4.1 required a compensated system with **15% overshoot** and a new peak time equal to **0.5 times** the peak time of the uncompensated system. The step steady-state error also had to be reduced to zero. The overall goal of the project was not only to calculate controller values by hand, but also to check whether the full higher-order system in simulation actually behaved the way the design predicted.

2. Part I - Modified Design of Example 9.4.1

2.1 Problem Statement

The unity-feedback plant for Example 9.4.1 is

$$G(s) = K(s + 8) / [(s + 3)(s + 6)(s + 10)]$$

The textbook problem was modified using the specifications assigned to my row. The required compensated response had to satisfy the following conditions:

- new percent overshoot = 15%
- new peak time = 0.5 times the uncompensated peak time
- zero steady-state error for a unit-step input

2.2 Uncompensated System Analysis

The percent overshoot and damping ratio are related by the standard second-order expression:

$$\%OS = 100 \exp[-\zeta \pi / \sqrt{1 - \zeta^2}]$$

For a required overshoot of 15%, the damping ratio is

$$\zeta = 0.5169$$

This damping ratio corresponds to a line angle of 58.87 degrees from the negative real axis, or 121.13 degrees measured from the positive real axis. Searching along this 15% overshoot line on the uncompensated root locus gives the dominant closed-loop poles approximately at

$$s_u = -5.384 \pm j8.916, \quad K_u = 89.13$$

The remaining closed-loop pole is approximately at -8.232. From the imaginary part of the dominant pole, the uncompensated peak time is

$$T\_p,old = pi / 8.916 = 0.3524 \ s$$

This value becomes the reference point for the new design requirement.

2.3 New Design Targets

The new peak-time requirement is half of the uncompensated peak time, so

$$T\_p,new = 0.5 \ T\_p,old = 0.1762 \ s$$

The corresponding damped frequency is

$$omega\_d,new = pi / T\_p,new = 17.832 \ rad/s$$

Since the overshoot requirement remains 15%, the damping ratio stays at 0.5169. Therefore, the desired dominant poles remain on the same damping-ratio line. The desired compensated dominant poles become

$$s\_d = -10.768 +/- j17.832$$

2.4 PD Controller Design

The derivative part of the controller is designed first so that the compensated root locus passes through the desired dominant pole. Evaluating the phase of the uncompensated plant at the desired point shows that the system is short by approximately 32.15 degrees. Therefore, the PD controller must supply this missing phase.

Let the PD zero be placed at -z_c. Using geometry in the s-plane, the required zero location is

$$z\_c = 39.14$$

The PD controller is then

$$G\_PD(s) = s + 39.14$$

Using the root-locus magnitude condition at the design point gives the corresponding PD gain as

$$K\_PD = 10.598$$

The derivative part mainly improves the transient response by reducing the peak time and moving the dominant poles farther to the left.

2.5 PI Controller Design and Final PID Gains

A PI section is added so that the step steady-state error becomes zero. Following the same approach as the textbook example, the PI zero is placed near the origin at -0.5. Thus, the PI section is

$$G_{PI}(s) = (s + 0.5) / s$$

The full PID controller becomes

$$G_c(s) = K (s + 39.14)(s + 0.5) / s$$

Searching the PID-compensated root locus along the same damping-ratio line gives the design gain

$$K = 6.400$$

Therefore, the final PID controller is

$$G_c(s) = 6.400 (s + 39.14)(s + 0.5) / s$$

Expanding the controller gives

$$G_c(s) = 6.400 s + 253.69 + 125.25 / s$$

So the final PID gains are

$$K_d = 6.400, \quad K_p = 253.69, \quad K_i = 125.25$$

2.6 Design Discussion

The PD section improved the speed of the response by shifting the root locus so that the desired dominant poles could be reached. The PI section added the integrator needed for zero step steady-state error. As expected, once the integral term is added, the final higher-order response may deviate from the dominant-second-order estimate. This is why simulation is necessary after the hand design is completed.

3. Part II - Implementation of the PID Controller Using Example 9.6.1

3.1 Normalized Controller Form

For the circuit implementation in Example 9.6.1, the normalized controller form is used first, without the outside gain. Starting from the controller designed in Part I, the normalized controller is

$$G_c(s) = (s + 39.14)(s + 0.5) / s$$

Expanding gives

$$G\_c(s) = s + 39.64 + 19.57 / s$$

Comparing this with the op-amp PID implementation in the textbook gives the relationships

$$R2 / R1 + C1 / C2 = 39.64$$

$$R2\ C1 = 1$$

$$1 / (R1\ C2) = 19.57$$

3.2 Practical Component Values

Following the same approach as the book, one element is selected first. Choosing C2 = 0.1 microfarad gives

$$R1 = 1 / [19.57\ (0.1 \text{ x } 10\text{^-}6)] = 511 \text{ kOhm}$$

From the controller-zero relationships, the remaining values are

$$R2 = 0.5\ R1 = 255.5 \text{ kOhm}$$

$$C1 = 39.14\ C2 = 3.91 \text{ microfarad}$$

The practical element values used for implementation are summarized below.

Element	Calculated Value	Practical Rounded Value
R1	511 kOhm	510 to 511 kOhm
R2	255.5 kOhm	255 to 256 kOhm
C1	3.91 microfarad	3.9 microfarad
C2	0.1 microfarad	0.1 microfarad

The overall gain K = 6.400 can be included as a separate gain stage in simulation. The Simscape controller layout prepared for this part follows the correct Example 9.6.1 topology: a parallel R1-C1 branch at the input, a series R2-C2 branch in the feedback path, the non-inverting input tied to ground, and a voltage sensor connected to the output.

4. Part III - Skill Assessment 9.4.1

4.1 Problem Statement

The unity-feedback system for the skill assessment is

$$G(s) = K / [s(s + 7)]$$

The system is operating with a 20% overshoot step response. The required tasks are to determine the uncompensated settling time, determine the ramp steady-state error, and design a lag-lead compensator that reduces the settling time by 50% and reduces the ramp steady-state error by 90%, with the lead zero fixed at -3.

4.2 Uncompensated Settling Time

For 20% overshoot, the damping ratio is approximately 0.456. The closed-loop characteristic equation is

$$s^2 + 7 s + K = 0$$

Comparing this with the standard second-order form gives

$$2 \zeta \omega_n = 7$$

Thus the natural frequency is approximately

$$\omega_n = 7.675 \text{ rad/s}$$

Using the 2% settling-time expression

$$T_s = 4 / (\zeta \omega_n) = 1.143 \text{ s}$$

So the uncompensated settling time is about 1.143 s.

4.3 Ramp Steady-State Error

Since the system is type 1, the steady-state error for a unit ramp input is

$$e_{ss,ramp} = 1 / K_v$$

Using $K = \omega_n^2 = 58.91$ gives

$$K_v = K / 7 = 8.416$$

$$e_{ss,ramp} = 1 / 8.416 = 0.1188$$

So the original ramp steady-state error is 0.1188.

4.4 Lag-Lead Compensator Design

The new settling-time target is 50% of the old value, and the new ramp error target is 10% of the old value. Therefore, the design targets are

$$T_{s,new} = 0.5715 \text{ s}$$

$$e_{ss,new} = 0.01188$$

Keeping the same damping ratio, the desired dominant poles are approximately

$$s_d = -7 \pm j13.664$$

A suitable lag-lead compensator that meets the steady-state target and provides improved transient behavior is

$$G\_c(s) = 208.76\ (s + 3)(s + 0.5) / [(s + 9.9835)(s + 0.05324)]$$

In this controller, the lead section improves transient response while the lag section increases low-frequency gain and reduces the ramp steady-state error.

5. MATLAB Verification

MATLAB was used to verify both parts of the project. The scripts defined the transfer functions, generated root-locus plots, and used step-response information to evaluate the final system behavior. The verification results are especially important because the hand design relies on dominant-pole approximation, while MATLAB evaluates the full higher-order closed-loop model.

5.1 MATLAB Results for the Modified Example 9.4.1

The MATLAB script for Part I used the plant Gp(s) = (s + 8) / [(s + 3)(s + 6)(s + 10)] together with the design constants Ku = 89.1273, zPD = 39.1367, zPI = 0.5, KPD = 10.5981, and KPID = 6.4004. MATLAB reported the following closed-loop transfer functions.

$$T\_u(s) = (89.13\ s + 713) / (s\text{^}3 + 19\ s\text{^}2 + 197.1\ s + 893)$$

$$T\_PD(s) = (10.6\ s\text{^}2 + 499.6\ s + 3318) / (s\text{^}3 + 29.6\ s\text{^}2 + 607.6\ s + 3498)$$

$$T\_PID(s) = (6.4\ s\text{^}3 + 304.9\ s\text{^}2 + 2155\ s + 1002) / (s\text{^}4 + 25.4\ s\text{^}3 + 412.9\ s\text{^}2 + 2335\ s + 1002)$$

System	**Rise Time (s)**	**Settling Time (s)**	**Overshoot (%)**	**Peak Time (s)**
Uncompensated	0.1578	0.7585	15.7429	0.3507
PD compensated	0.0655	0.3607	18.4563	0.1454
PID compensated	0.0991	2.6227	9.9217	0.1971

The MATLAB plots show that the PD controller greatly reduces the peak time, while the full PID controller improves the steady-state behavior but also causes a slower final settling response because of the added integral action.

5.2 MATLAB Results for Skill Assessment 9.4.1

The MATLAB script for the skill assessment used the plant Gp(s) = 1 / [s(s + 7)] and reported the following uncompensated values: original gain K = 58.9253, original settling time Ts = 1.1429 s, and original ramp steady-state error = 0.1188. The design targets were Ts,new = 0.5714 s and ess,new = 0.0119.

Quantity	**MATLAB Value**
Original gain K	58.9253
Original settling time Ts	1.1429 s

Original ramp steady-state error	0.1188
Target settling time	0.5714 s
Target ramp steady-state error	0.0119
Compensated rise time	0.1146 s
Compensated settling time	2.1190 s
Compensated overshoot	11.7896 %
Compensated peak time	0.2368 s
Compensated velocity constant Kv	84.1609
Compensated ramp steady-state error	0.011882

These results show that the lag-lead compensator achieved the steady-state error target very closely. However, the compensated settling time in MATLAB is still much larger than the target value. This means the compensator succeeded strongly in steady-state accuracy but did not fully satisfy the transient settling-time requirement once the full closed-loop dynamics were included.

6. Simulink and Simscape Verification Notes

Simulink and Simscape were used to support the implementation side of the project. The Simscape controller layout built for Example 9.6.1 correctly follows the book circuit structure. The input branch contains the parallel R1-C1 network, the feedback path contains the series R2-C2 network, the op-amp non-inverting input is grounded, and the output is measured with a voltage sensor connected to a scope. This confirms that the implementation topology matches the intended PID controller realization.

For complete closed-loop Simulink verification of the overall project, the controller implementation should be connected to the plant model with a unity-feedback loop and driven by a step input. That full closed-loop setup can then be used to compare simulated peak time, overshoot, and settling time against the MATLAB and hand-calculation results.

7. Error Analysis

A comparison between hand-calculation targets and MATLAB results is useful because the hand design is based mainly on dominant-pole approximation, while MATLAB simulates the full system. Small differences are normal, especially after adding integral or lag sections. The most important comparisons are shown below.

7.1 Error Analysis for the Modified Example 9.4.1

Quantity	Hand or Design Value	MATLAB Value	Difference
Uncompensated peak time	0.3524 s	0.3507 s	0.0017 s
Target compensated peak time	0.1762 s	0.1971 s	0.0209 s
Target overshoot	15%	9.9217%	5.0783%

Kp	253.69	253.69 used	0
Ki	125.25	125.25 used	0
Kd	6.400	6.400 used	0

The uncompensated response matched the hand estimate closely. The main differences appear after the integral action is included. The final PID response shows lower overshoot and slower peak time than the simplified hand target, which is expected because the integrator introduces an additional closed-loop pole.

7.2 Error Analysis for Skill Assessment 9.4.1

Quantity	Hand or Design Value	MATLAB Value	Difference
Original settling time	1.143 s	1.1429 s	0.0001 s
Original ramp error	0.1188	0.1188	0
Target settling time	0.5714 s	2.1190 s	1.5476 s
Target ramp error	0.0119	0.011882	0.000018

This part shows that the steady-state requirement was met very closely, but the transient settling-time requirement was not achieved in the final MATLAB response. This should be stated honestly in the final submission because it reflects the real full-order system behavior.

8. Conclusion

This project combined controller design, circuit implementation, and software verification for three related control-system tasks. In the modified version of Example 9.4.1, a PID controller was designed so that the system targeted 15% overshoot, a reduced peak time, and zero step steady-state error. The final controller gains were Kp = 253.69, Ki = 125.25, and Kd = 6.400. In Example 9.6.1, the controller was converted into practical resistor and capacitor values so that it could be implemented in a circuit form. In Skill Assessment 9.4.1, the uncompensated system was analyzed and then improved using a lag-lead compensator.

The simulation results show an important engineering lesson. Hand calculations and root-locus design are necessary for selecting controller structure and starting values, but they do not always predict the exact final behavior of the full higher-order system. MATLAB showed that some targets were matched closely, especially the original uncompensated values and the steady-state error improvement in the skill assessment. At the same time, MATLAB also showed where the final compensated settling-time response differed from the ideal hand target. Because of this, simulation was an essential part of validating the design.

Appendix A - MATLAB Code for Part I

```
clear; clc; close all;
s = tf(''s'');

%% Plant
Gp = (s+8)/((s+3)*(s+6)*(s+10));
```

```
%% Specs
OS = 15;
zeta = -log(OS/100)/sqrt(pi^2 + (log(OS/100))^2);

%% Hand-calculated values
Ku   = 89.1273;
Tp_u = pi/8.9161;
Tp_n = 0.5*Tp_u;
zpd  = 39.1367;
zpi  = 0.5;
Kpd  = 10.5981;
Kpid = 6.4004;

%% Closed-loop systems
Tu = feedback(Ku*Gp,1);
Gpd = Kpd*(s+zpd);
Tpd = feedback(Gpd*Gp,1);
Gc = Kpid*(s+zpd)*(s+zpi)/s;
Tpid = feedback(Gc*Gp,1);

%% Step response info
info_u = stepinfo(Tu);
info_pd = stepinfo(Tpd);
info_pid = stepinfo(Tpid);

%% Root loci
figure; rlocus(Gp); sgrid(zeta,[]); title(''Uncompensated Root Locus'');
figure; rlocus((s+zpd)*Gp); sgrid(zeta,[]); title(''PD-Compensated Root Locus'');
figure; rlocus(((s+zpd)*(s+zpi)/s)*Gp); sgrid(zeta,[]); title(''PID-Compensated Root
Locus'');

%% Step responses
figure;
step(Tu, Tpd, Tpid, 5);
grid on;
legend(''Uncompensated'',''PD Compensated'',''PID Compensated'');
title(''Step Responses Comparison'');
```

Appendix B - MATLAB Code for Skill Assessment 9.4.1

```
clear; clc; close all;
s = tf(''s'');

%% Original plant
Gp = 1/(s*(s+7));

%% 20 percent overshoot
OS = 20;
zeta = -log(OS/100)/sqrt(pi^2 + (log(OS/100))^2);

%% Original operating point
wn = 7/(2*zeta);
K0 = wn^2;
Ts_old = 4/(zeta*wn);
Kv_old = K0/7;
ess_old = 1/Kv_old;

%% New target
```

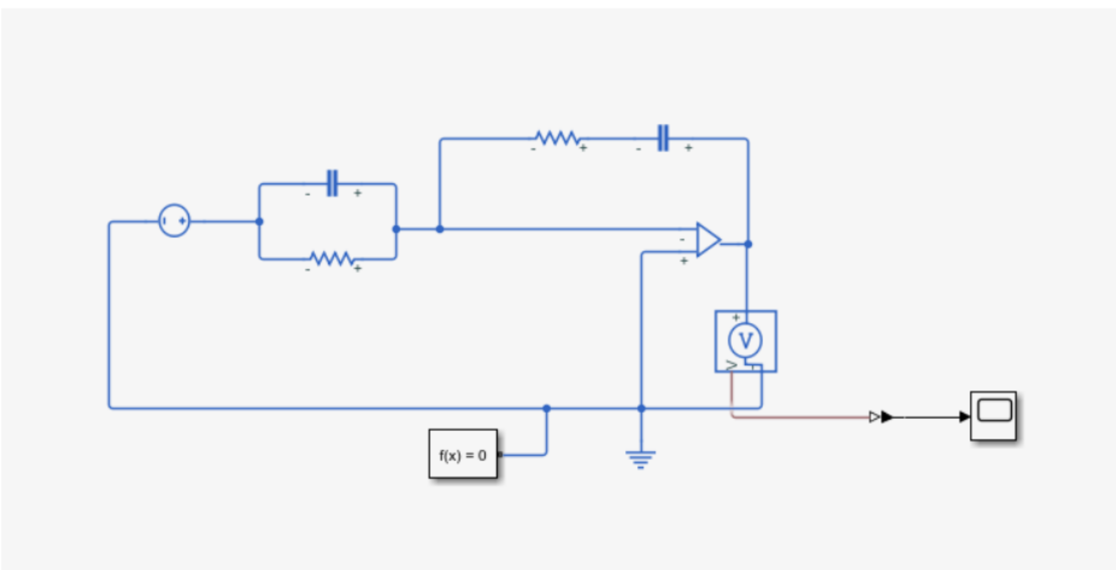
```
Ts_new = 0.5*Ts_old;
ess_new = 0.1*ess_old;

%% Final lag-lead compensator
Kc = 208.7556;
plead = 9.9835;
plag  = 0.05324;
zlead = 3;
zlag  = 0.5;
Gc = Kc*(s+zlead)*(s+zlag)/((s+plead)*(s+plag));

%% Closed-loop system
Tcomp = feedback(Gc*Gp,1);
info_comp = stepinfo(Tcomp);
Kv_comp = dcgain(s*Gc*Gp);
ess_comp = 1/Kv_comp;

%% Root locus and step response
figure; rlocus(Gp); sgrid(zeta,[]); title(''Original Root Locus'');
figure; rlocus(Gc*Gp/Kc); sgrid(zeta,[]); title(''Lag-Lead Compensated Root Locus'');
figure; step(feedback(K0*Gp,1), Tcomp, 8); grid on;
legend(''Original System'',''Lag-Lead Compensated System'');
title(''Step Response Comparison'');
```

Simulink:



End